5. Signal-to-Quantization Noise Ratio (SQNR) Analysis

Simulate a full-scale sinusoidal signal using different word lengths and analyze the effect of quantization on the Signal-to-Quantization Noise Ratio (SQNR). (25) (BL5) (WK4)

Given Data:

- Input Signal = Full-scale sinusoid
- Sampling Frequency = 8 kHz
- Word Lengths = 4, 6, 8, 10, 12, 16 bits

Tasks:

1. Generate a full-scale sinusoidal signal.
2. Quantize the signal using the specified word lengths.
3. Compute the signal power.
4. Compute the quantization noise power.
5. Calculate the SQNR for each word length.
6. Compare the simulated SQNR values with the theoretical SQNR values and discuss the results.

Answers:

#CODE

```
%% Signal-to-Quantization Noise Ratio (SQNR) Analysis
% Tasks:
% 1. Generate a full-scale sinusoidal signal.
% 2. Quantize the signal using specified word lengths (4, 6, 8, 10, 12, 16 bits).
% 3. Compute signal power.
% 4. Compute quantization noise power.
% 5. Calculate SQNR for each word length.
% 6. Compare simulated vs. theoretical SQNR values.
clear; clc; close all;
```

```
%% 1. Given Data & Simulation Setup

fs = 8000;              % Sampling Frequency = 8 kHz

fm = 100;               % Signal Frequency = 100 Hz

t_final = 0.05;         % Duration = 50 ms

t = 0 : 1/fs : t_final - (1/fs); % Time vector

A = 1.0;                % Full-scale peak amplitude

word_lengths = [4, 6, 8, 10, 12, 16]; % Word lengths in bits

% Generate full-scale sinusoidal signal

x = A * sin(2 * pi * fm * t);

%% 2. Compute Input Signal Power

% Power of full-scale sinusoid: P_s = mean(x^2)

P_s = mean(x.^2);

%% 3. Quantization and SQNR Computation

num_bits = length(word_lengths);

P_n = zeros(1, num_bits);

sqnr_sim_db = zeros(1, num_bits);

sqnr_theo_db = zeros(1, num_bits);

fprintf('==========================================================================\n'
);

fprintf(' %-10s | %-12s | %-18s | %-18s \n', 'Bits (B)', 'Step Size', 'Theo SQNR (dB)', 'Sim SQNR (dB)');

fprintf('==========================================================================\n'
);

for i = 1:num_bits

    B = word_lengths(i);

    L = 2^B;            % Number of quantization levels

    delta = (2 * A) / L; % Quantization step size


    % Mid-rise uniform quantization

    % Quantizes continuous signal into levels centered in intervals
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```
    x_quant = delta * (floor(x / delta) + 0.5);

    % Quantization Error / Noise
    e = x - x_quant;

    % Quantization Noise Power
    P_n(i) = mean(e.^2);

    % Simulated SQNR in dB
    sqnr_sim_db(i) = 10 * log10(P_s / P_n(i));

    % Theoretical SQNR in dB: 6.02 * B + 1.76
    sqnr_theo_db(i) = 6.02 * B + 1.76;

    fprintf(' %-10d | %-12.6f | %-18.2f | %-18.2f \n', ...
            B, delta, sqnr_theo_db(i), sqnr_sim_db(i));
end
fprintf('==========================================================================\n\
n');
%% 4. Plotting Results
% Plot 1: SQNR vs Word Length
figure('Name', 'SQNR Performance Analysis', 'NumberTitle', 'off');
plot(word_lengths, sqnr_theo_db, 'r--o', 'LineWidth', 2, 'MarkerSize', 8); hold on;
plot(word_lengths, sqnr_sim_db, 'b-s', 'LineWidth', 2, 'MarkerSize', 8);
grid on;
xlabel('Word Length B (bits)', 'FontSize', 12);
ylabel('SQNR (dB)', 'FontSize', 12);
title('Theoretical vs. Simulated SQNR', 'FontSize', 14);
```

```
legend('Theoretical (6.02B + 1.76 dB)', 'Simulated SQNR', 'Location', 'northwest');

% Plot 2: Original Signal vs Quantized Signals (Example: 4-bit vs 8-bit)

figure('Name', 'Quantization Waveforms', 'NumberTitle', 'off');

% 4-bit Quantization Waveform

delta_4 = (2 * A) / (2^4);

x_q4 = delta_4 * (floor(x / delta_4) + 0.5);

% 8-bit Quantization Waveform

delta_8 = (2 * A) / (2^8);

x_q8 = delta_8 * (floor(x / delta_8) + 0.5);

subplot(2,1,1);

plot(t * 1000, x, 'k', 'LineWidth', 1.5); hold on;

stepplot_4 = plot(t * 1000, x_q4, 'r', 'LineWidth', 1.2);

grid on;

xlabel('Time (ms)'); ylabel('Amplitude');

title('4-bit Uniform Quantization');

legend('Original Signal', '4-bit Quantized');

subplot(2,1,2);

plot(t * 1000, x, 'k', 'LineWidth', 1.5); hold on;

plot(t * 1000, x_q8, 'b', 'LineWidth', 1.2);

grid on;

xlabel('Time (ms)'); ylabel('Amplitude');

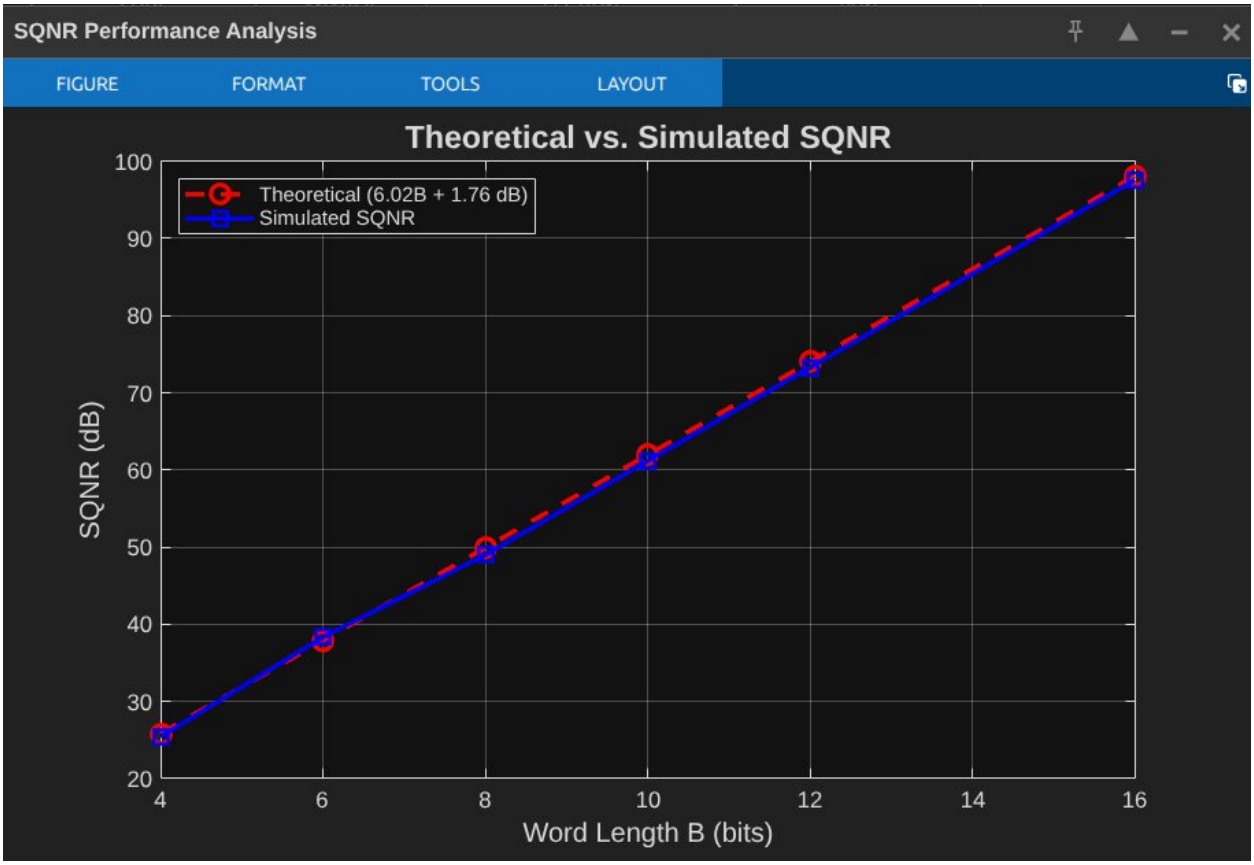
title('8-bit Uniform Quantization');

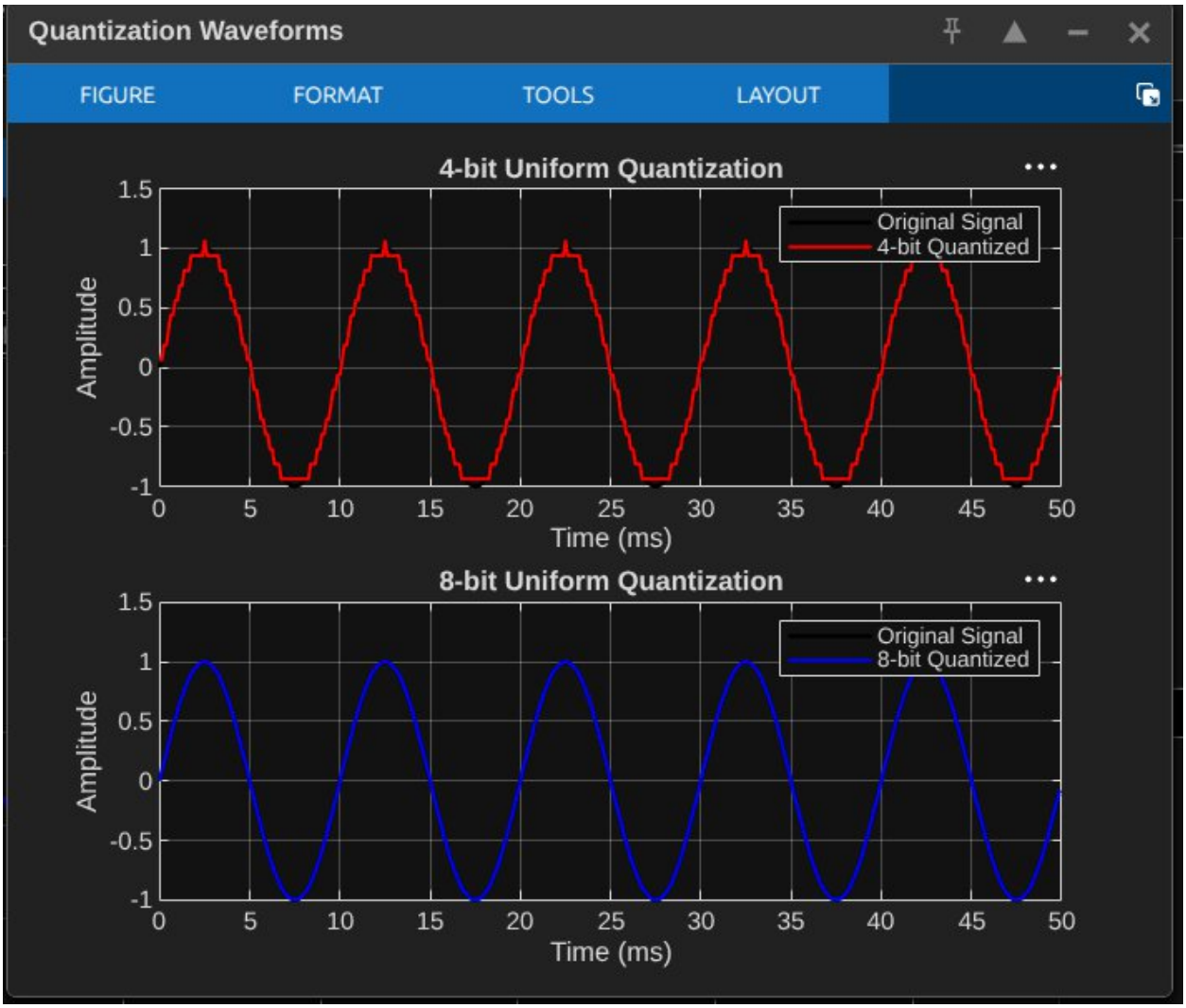
legend('Original Signal', '8-bit Quantized');
```

Output:

Command Window

Bits (B)	Step Size	Theo SQNR (dB)	Sim SQNR (dB)
4	0.125000	25.84	25.42
6	0.031250	37.88	38.33
8	0.007812	49.92	48.99
10	0.001953	61.96	61.17
12	0.000488	74.00	73.27
16	0.000031	98.08	97.56




Quantization Waveforms
FIGURE
FORMAT
TOOLS
LAYOUT
4-bit Uniform Quantization
Original Signal
4-bit Quantized
Amplitude
Time (ms)
8-bit Uniform Quantization
Original Signal
8-bit Quantized
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